



BSc (Hons) in **Computer Science**
SE3062 : Intelligent Systems
 Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 01

Objective & Lecture Mapping: Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

Part 1: Code Analysis & PEAS Evaluation

Task 0 : Setting up and getting the starter Code

1. Go to the [Git Hub Repo](#). Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" brach of the repo.

Task 1.1: The Environment State (`__init__`)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

e - environment
 a -actuators
 s - sensors

2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

`self.width` and `self.height` - the size of the grid
`self.walls` - represents obstacles in the environment
`self.food_positions` - represent the locations of the food items

3. (Analyze) Based on the variables initialized (specifically `self.opponents`), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

multi agent
 the environment includes the main agent and one or more opponent agents (`self.opponents`)

Task 1.2: The Perception Subsystem (`get_percept`)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

the entire sequence of observations and agent has received over time from its environment

5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

sensors

6. (Evaluate) Based on the exact dictionary returned by `get_percept()`, is this environment "Fully Observable" or "Partially Observable"? Explain why based on what the agent can and cannot see.

fully observable
the agent can access the complete current state needed to make decisions.

Task 1.3: Action Execution (`execute_action`)

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

performance measure

8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

the performance measure should judge what the agent is achieving, now how the agent is thinking.

Part 2: Guided Modification & Deep Theory Mapping

Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

Step 2.1: Extending the Environment Initialization (`__init__`)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples safely avoiding position `(0, 0)`, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally **hide** this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

the `self.toxic_traps` are added to the environment but hidden from the agent's sensors. so it becomes partially observable

Step 2.2: Updating the Perception Subsystem (`get_percept`)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos) in self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding 'smells_toxin' , you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

it improves the agent's rationality because it gives the agent additional information to make better decisions

Step 2.3: Modifying Action Execution & Visual Rendering (execute_action & draw_grid)

1. **Implementation Instructions:** In `execute_action` , check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score` by 15 points. In `draw_grid` , iterate through traps to render custom purple shapes on the canvas.

11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?

the faulty performance metric can make an agent optimize the score instead of the real goal. a vacuum agent could repeatedly clean an already clean area to gain rewards, without actually making the area cleaner

Finalize and Lock Lab 01 Analytical Evaluation



FACULTY OF COMPUTING